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**Investment decision making under
uncertainty based on internet
recommendations**

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Abstract

The main objective of this bachelor thesis is to make investment portfolios based on recommendations from Internet news servers. Due to high inflation, there is a growing interest in investing even among less involved people, who often follow such suggestions. Firstly, articles are analyzed, and investment instruments are selected. Next, based on the methodology, a statistical overview of historical data is shown, and the simulated values of portfolio returns and risks are generated according to the relevant probability distribution, taking into account regular investment. The resulting parameters are used to optimize a portfolio. The portfolio is composed of disparate investment instruments, such as investment into water, gold, agricultural land, stocks of CEZ company, and stocks of a fond following the S&P 500 index. The results show that online articles provide a helpful overview but not appropriate recommendations. According to news portals and optimization, investing in agricultural land is the most advantageous.

Keywords

internet news server, investment decision making, Markowitz model, Monte Carlo simulation, portfolio optimization

Abstrakt

Hlavním cílem této bakalářské práce je vytvořit investiční portfolia na základě doporučení internetových zpravodajských serverů. Kvůli vysoké inflaci roste zájem o investování i mezi méně zainteresovanými lidmi, kteří se takovými návrhy často řídí. Nejprve jsou analyzovány články a vybrány investiční nástroje. Dále je na základě metodiky vytvořen statistický přehled historických dat a simulované hodnoty výnosů a rizik portfolia jsou generovány podle příslušného pravděpodobnostního rozdělení, při zohlednění pravidelné investice. Výsledné parametry jsou použity pro optimalizaci portfolia. Portfolio je složeno z nesusoudných investičních instrumentů, konkrétně investice do vody, zlata, zemědělské půdy akcií společnosti ČEZ a akcií fondu sledujícího index S&P 500. Výsledky ukazují, že internetové články poskytují užitečný přehled, ne však vhodná doporučení. Nejvíce se dle článků a optimalizace vyplatí investovat do zemědělské půdy.

Klíčová slova

internetový zpravodajský server, investiční rozhodování, Markowitzův model, Monte Carlo simulace, optimalizace portfolia

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Introduction

People tend to read the news more online than buy the print in today's world. We want new information delivered quickly every time we ask for it. In this fast world, we also look for fast solutions for everything. At the same time, we live in a time of high inflation, and we need to think about money allocation. It is mentioned all around us how savings in our accounts are shrinking, and therefore the topic of investment is brought up. Since people are often unfamiliar with this subject, they seek advice on investing money in internet news portals. However, recommendations on such servers are often not based or shown on numbers or scientific research. They are more about a person's experience, especially if working in the financial market. Hence, the author has decided to analyze these articles that recommend different types of investments. The reports regarding this theme that are 5-10 years old (therefore written in the years 2012-2017) will be read, and the financial instruments that were the most recommended to invest in will be selected. Next, financial instruments will be analyzed, how their price developed in time, and whether the investment would have been worth it. Furthermore, this thesis will show an investment model to determine the percentage of invested amount for each financial instrument in the portfolio.

As to the objectives and expected benefits, this thesis aims to verify financial advice in internet news articles mainly based on personal experience and gut rather than on calculated proof. Therefore, thanks to the quantitative methods applied in this thesis, it should be determined whether following the advice of internet news services would have been worthwhile. Simultaneously, the motivation is also to decide what a portfolio constructed according to such recommendations should look like. In other words, this thesis will provide the best distribution of financial assets to the selected instruments since that is also mostly lacking in mentioned articles. Without a recommendation on the specific allocation of money, it can be challenging for an uninvolved person to make an investment decision. With all mentioned above, the reader must bear in mind that the financial world is unpredictable, and all calculated results are based on the past, which does not have to imply how the future will behave.

The goal will be achieved by first collecting the needed time-series data. Afterward, various analyses will be conducted based on the problem-solving methodology discussed in the following chapters of this thesis. For example, statistical computations will take place, such as regression of prices on time and fitting distributions to the data of each selected financial instrument. According to the articles described below, a stochastic model for financial optimization will be created. This model is to be finding the best distribution of invested money between the financial instruments selected in the theoretical part of this thesis while lowering the risk of the investment and maximizing the return on equity. At the same time, the return will be shown as stochastic, as it will be taken as a random variable with a specified probability distribution. As a result, returns will be generated, from which the expected return and risk characteristics will be calculated. Therefore, there will not be a determinate result.

Instead, the outcome will be a mean of multiple optimization runs. Since the information in internet newspaper articles is diverse, this paper includes non-traditional investment instruments. Examples are an investment in agricultural land or investment in drinking water. For this reason, it will be more challenging to find appropriate procedures for analysis, as the assets to be invested in will be hard to compare and combine into one portfolio.

Regarding the thesis structure, firstly, a theoretical search will determine what other articles, papers, and books were already written about this topic to ensure this thesis brings something new and innovative. After, lay articles about investing recommendations will be analyzed, and the most frequently mentioned financial instruments will be selected. The aim is to pick approximately four to six tools to create a portfolio. Secondly, the selected financial instruments will be analyzed so that the reader will have a comprehensive overview of each tool and its development in time. Thirdly, a mathematical model is to be created. This model will determine the structure of the best possible portfolio for an investor. Lastly, a conclusion will be made, summarizing the development of this thesis. The outcome will include a recommendation of a portfolio structure based on optimization results.

1. Literature review

The first chapter of this thesis deals with the analysis of articles already written on a similar topic, a summary of the methods used and the results already achieved. For the purposes of this paper, it is necessary to take into account the progress in portfolio optimization under conditions of uncertainty, which in this case leads to the Monte Carlo simulation. The aim is to analyze articles in online news servers, so the methodology of doing so is also summarized.

1.1 Portfolio theory

To begin, portfolio theory will be addressed. The first important piece of information is that we work with uncertainty when investing. This uncertainty cannot be eliminated. If an investor was confident in the future, one would only invest in the commodity with the highest return. Alternatively, if the instruments had precisely the same rates of return, any combination of them would be accepted. However, one would certainly not choose lower-yielding investments for the sake of portfolio diversification. Additionally, while composing a portfolio, different approaches to uncertainty can be taken into account [29, 31, 32].

The best-known model is probably the Mean-Variance Model, published in 1952 by Harry Markowitz. Here, the choice should be guided by the maximum expected utility. The investor chooses from the efficient frontier of the mean-variance, approximately maximizing expected utility for a wide range of concave (risk-averse) utility functions [30]. High expected return is a desirable thing for the investor, and, on the other hand, the variance of return is considered undesirable. At the same time, a rule of thumb suggests that an investor should diversify his funds while maximizing expected returns. Thus, an investor diversifies ones funds so that the invested in securities provide the maximum expected return. Risk is, in this case, taken as variance or standard deviation of expected returns. This statistical measure of risk makes the model relatively easy to apply and interpret, since it is an essential statistical variable. But with simplicity comes a downside. Specifically, there is a problem with penalizing positive deviations, meaning that we also minimize the deviation in the form of expected return growth. Another major disadvantage is that this model assumes normality of yields of financial assets, which in most cases is not met [29].

Next discussed was for example the Mean-Semivariance model. Semivariance is an alternative way to measure risk. The model penalizes only negative changes [32]. Most often, the standard deviation accounts only for fluctuations below the mean. This model is also called Downside risk [17]. Alternatively, there is a concept of average absolute negative deviation. This concept calculates with estimated returns below the mean, and, conversely, positive growth remains unaccounted for in the variability measure [8]. Over the years, many other concepts have been developed to optimize a portfolio, such as the Mean-Absolute Deviation model or Conditional Value at Risk model [22]. It is also possible to measure the average

Probability of loss, where it is calculated what percentage of the time the investor is at loss [45]. As the main alternatives for measuring the risk to the variance and semivariance H. Markowitz considers Expected loss, Expected absolute deviation, Probability of loss, and Maximum loss. The determination of the rate of return must also be considered. For example, it is possible to measure return as a geometric mean. This procedure is recommended if the risk is expressed as a semivariance, as the lack of a negative attitude to risk at abnormally high returns can be a problem here. At the same time, it should be bear in mind that this approach to returns is more computationally intensive [33].

Other incentives should also be included when compiling the portfolio. Different conditions are usually incorporated in the model. For example, the number of investment instruments in the final portfolio fundamentally determines its diversification. Basically, the more the amount invested is divided among the different instruments, the more diversified the portfolio is but approximately 30 investment assets are considered sufficient. Diversification reduces the risk of the portfolio as a whole. In addition, there is also the issue of correlation between instruments; specifically, there should be some negatively correlated variables in the portfolio. This way, fluctuations in value are better smoothed out [47].

Lastly, it is necessary to mention multi-criteria models that deal with complex decisions that often need to be assessed from multiple perspectives. In multi-criteria decision-making, aspects that should be minimized usually need to be combined with those that should be maximized. Therefore, it is necessary to unify the criteria so that they are comparable. Also, it is essential to decide whether one aspect is more relevant than another, if yes, weights should be assigned. In portfolio optimization, a multi-criteria decision-making approach is quite common, as the investor usually requires the highest possible return at the lowest possible risk. The return is thus maximized in the optimization and the risk is minimized. Such a model could also include, for example, the liquidity aspect, which would again be a maximization criterion [15].

1.2 Monte Carlo simulation

As already mentioned, uncertainty plays a significant role in the investment world. Experts try to reduce this uncertainty as much as possible to make their decisions future-proof to the full. This is why the Monte Carlo method is applied in portfolio analysis. In this method, randomness is used, and then it is taken in a determinate way. This idea was very revolutionary. It was introduced by the French scientist Georges-Louis Leclerc, Comte de Buffon, who lived in the eighteenth century. His most famous study is probably one called Buffon's needle. Georges-Louis Leclerc, Comte de Buffon, threw a needle on a wooden ground, trying to find out the probability of the needle falling so that it lies across the gap between the two boards to estimate π . Interestingly, the Monte Carlo method was only a by-product of this experiment. However, this simulation deals with deterministic problems. Modern simulations solve the problem probabilistically, so called stochastically. This simulation-style

was first used in a Manhattan project when the Americans were trying to develop nuclear weapons in World War II. They named the method after the Monte Carlo casino located in Monaco [21].

Monte Carlo simulation focuses on the behavior of a variable under random sampling. Thus, the process of randomly drawing samples is repeated, and their behavior is observed. These repeats create a sort of pseudo-population, and if there are enough samples, it can be argued that the population itself would behave according to the pseudo-population. While conducting the Monte Carlo simulation, it is needed to specify the pseudo-population, so it can be used to generate samples, create an algorithm, sample from the pseudo-population, save results to a vector, and repeat trials. The relative frequency of the results is produced, and, lastly, the results are interpreted [35].

For Monte Carlo simulation, a probability distribution is used. However, problems can arise, as determining the distribution may not always be easy. The variable element is then expressed as a random variable with a specified probability distribution [7, 8]. The accuracy of the estimated parameters depends on the number of times the Monte Carlo simulation is repeated. One thousand simulations could be considered a sufficient number [8, 14].

There are also some important issues to consider when using this method. For example, what our key questions are, what outcomes we expect, and in what format we want the results. What these outputs will be used for or, for example, how accurate do these results need to be. It is also important to check the quality of the inputs to ensure that the method works correctly [21].

1.3 Article analysis and investment recommendations

For the purposes of this thesis, it is necessary to analyze lay articles from internet news portals. Therefore, it is essential to understand article analysis. Articles written as expert recommendations are often written in natural language, which in many cases is meant to express and evoke human emotion. Some emotions may be subtly hidden in the text, such as the evocation of uncertainty, which occurs in small doses throughout the text. Thus, they enter the reader's subconscious without him/her noticing. On the other hand, the emotions in the article can be pronounced and confident of persuading the reader while admitting the matter. Knowing all types of feelings hidden in an article will show the actual intention with which it was written and what message it is meant to convey. There are two types of statements. Specifically, the objective and the subjective stance. There is a clear opinion or belief of an individual, group, or organization about a particular topic in a subjective stance. Objective statements merely show factual information. However, even facts can arouse emotions in the reader [3].

While reading an article, one has to wonder why the article was written in the first place. Whether it is a sophisticated piece that sticks to the primary idea and seeks an answer to a question or, on the contrary, just responds to the demand and writes about a popular topic among customers to attract attention and make money. Alternatively, whether it even functions as advertising [51].

In terms of investment recommendations, the vast majority of the articles did not propose a specific recommendation on where to invest. Nor were exact suggestions found in the years selected to analyze the internet articles in the next chapter, i.e. 2011 to 2017, or in the following years. This also suggests that such advice may not always be mathematically provable and fact-based. Existing advice is often formed through industry experience and market observation. The evidence on data is rather an exception. While it is undoubtedly possible to at least some extent, predict the development of commodity prices, as already mentioned, this is not a deterministic estimate, as the future is unknown. In addition, a lot depends on how much money each person wants to invest, whether they have any time frame for the investment, and their attitude to risk. Therefore, there is no one clear answer to where to put one's money [16]. Another problem is that some advising materials may not be from independent sources. Thus, some estimates may be exaggerated [5].

But the fact is that people do seek financial advice. Social media with investment content is growing in popularity. Most of the time, these sites let their users discuss investment opinions, and it is not easy to follow the advice, especially if one is not an expert on the topic. People nowadays also rely on positive or negative reviews. However, it should be kept in mind that there are also misleading or illogical contributions to evaluations. So the question is, to what extent to use people's wisdom and experience and to what extent to act on data and numbers. However, as mentioned several times in this chapter, even portfolio construction based on calculations is done under uncertainty. When making a decision, it is crucial to obtain information about the author of the opinion, the content, and form of the message, and the characteristics of the recommended investment [50].

2. Data

The choice of investment instruments is very important in this thesis as it is based on investment recommendations from internet portals. Therefore, a large part of this chapter focused on data is devoted to the analysis of these suggestions and then properties of the obtained data are described and compared.

2.1 Selection of portfolio components

Regarding the topic of this thesis, a model investor, a Czech citizen, is assumed. This investor does not have an overview of economic markets but has some financial surplus and is interested in allocating the money to appreciate. Therefore such an investor has access only to freely available information. At the same time, nowadays, people want easy to find, quick answers. Therefore search mainly on the internet on sites they know or are easy to understand. That is why the author chose to analyze internet news portals. It is often possible to come across recommendations about where to invest in these portals. The motivation for selecting portfolio components in this thesis comes from these suggestions. The aim was to select 4–6 mostly mentioned assets. Even though a diversified portfolio, as mentioned in Chapter 1, should be composed of at least 30 stocks, this paper focuses on the recommendations. That many assets are not likely to be mentioned in news articles repeatedly.

2.2 Suggestions from the internet news portals

Research of online articles was conducted, focusing on the ones published in the years 2011 to 2017. These years were selected for several reasons, mainly because before the year 2011, there were fewer online sources on this topic available to this date. At the same time, that gives us 5–10 years of price development, providing enough data for the analysis.

Firstly, it is essential to mention suggested ways to invest. Experts talking to the press mostly recommend regular investment. As an example, in an article on idnes.cz [41], it is stated that one of the clear advantages of regular investing is the lower risk of incorrect investment timing. When the investment is sent, for instance, every month, the purchase price of shares over time averages. Therefore shopping happens during both market growth and decline. Another advantage of this method is that the investor does not significantly limit their liquidity since the contribution to the funds is from a regular income with a relatively small amount. That is why such investment is considered better, especially for a person with an average wage and knowledge of the topic. Another article, this on aktualne.cz [18], focuses on the same topic. Similarly, the main identified advantage of regular investment is the averaging method. Periods of growth and decline are used to an advantage. In the event

of growth, the invested funds are appreciated. On the contrary, in the period of a decline, investors buy more securities for the invested funds. This article on [aktualne.cz](#) seems credible to the reader, but at the same time, it is a commercial article. Only the information on regular investments shown on the data is used in this case, as the author finds it unbiased. But many similar articles have been found with specific investment recommendations. They were not included in this work, as the author does not consider them impartial. Nonetheless, they can influence the readers of these portals.

Subsequently, the composition of the portfolio itself was analyzed. The most mentioned commodity was certainly gold, both in a positive and negative sense. For example, in an article from [investree.cz](#) [58], there are different views and approaches on gold. According to the writer, it is good to own some physical metals in the long run. At the same time, however, a higher yield cannot be expected. Gold was also mentioned in an article on [lidovsky.cz](#) from 2011 [1]. Here they state that the gold rush caught the whole world that year. Investing in gold should be a safe way to allocate money, but only if it is a long-term investment. Based on this source, contrary to popular belief, even gold is not an absolute certainty, but analysts expect that if the economic situation in the world fails to calm down, the price of gold will rise. That is always the case in crises.

The second most commonly recommended asset is real estate investing. But here, we cannot assume that our model investor can afford real estate, so this work will rather focus on purchasing land. Specifically, it was recommended to invest in agricultural land in internet sources. For example, in the already mentioned article from [lidovsky.cz](#) [1], it is written that investing in land can often be smarter than buying a house or an apartment. Especially when it comes to longer-term investments, when one waits five or more years for money. In another article from [lidovsky.cz](#) [9], it is mentioned that the most frequently recommended "refuge" for savings are certainly apartments, houses and land. This statement was also confirmed by Tomáš Končický, the head of an Austrian private bank, with his statement that in bad times real estate investment represents a sure bet.

Another mentioned investment in the Czech Republic was to allocate money to local companies, mainly to CEZ, a.s, which is one of the most important producers, distributors and traders of electricity, natural gas and heat in the Czech Republic [46]. However, these recommendations occur most often until 2012 and are practically no longer to be found. This may be related to the large drop in CEZ share prices in these years. In [aktualne.cz](#) in the year 2011 [24], the recommendation is to invest mainly in one's state because it is more possible to estimate whether the company is doing well. Indestructible companies are said to sell their products in any economic situation e.g. power plants and energy distribution companies, food companies and medical companies. According to this article, CEZ is a classic representative of such an unbeatable company in the Czech Republic. There is no physical principle on the horizon to replace electricity, so the company is set to have sales for decades.

Overall, there is a constant perception that the reader invests in international stocks as well. However, that is not a very clear instruction since there is a large selection of such investments. So, there are three most recurring ones in the portfolio for this paper. The S&P 500 index is known worldwide and often referred to for comparison in articles, for example on [idnes.cz](#), [aktualne.cz](#) or [econom.cz](#). [Idnes.cz](#) [41] used this comparison for example in the mentioned article about regular or one-time investment, where the differences were presented on the S&P 500 index. In [econom.cz](#) [53] it was mentioned that it is no longer the case that investments in the shares of the world's largest energy companies, for example, fond following the S&P 500 index, are a guarantee of profitability. Lastly, [aktualne.cz](#) [42] used a reference to the S&P 500 index directly in the title of an article on the fall of shares, where they stated that the S&P 500 is the lowest since October 2017 and that the decline in the price of US shares continues. This is the only instrument in the portfolio that was not explicitly recommended for the reader, rather the opposite, but was often mentioned as a base line or a standard. Therefore the author considers interesting to discover whether this investment will be included in the optimized portfolio in the Empirical Research chapter. Specifically, the asset included will be a stock mapping this index.

Last but not least included instrument is the investment into water, explicitly drinking water. An article on [investree.cz](#) [58] states that the population grows by about 80 million people a year. That would mean that in 23 years, the number of people would exceed nine billion. Our standard of living is also rising and people are buying more bottled water. It can also be assumed that water quality will be regulated and the problem of global warming will be addressed more. According to this article, everyone is certainly aware that water is a necessary commodity for life, which plays an increasingly important role in the future and it is certainly worth investing in it. According to an article from [investniweb.cz](#) [27], world demand for water has quadrupled in the past century. Additionally, its growth rate is not slowing down. On [novinky.cz](#) [48] it was posted that no other commodity has become more expensive in the Czech Republic since the early 1990s as tap water.

There have been many more ways to value your money mentioned in the online articles. Some experts have elegantly avoided the question of where readers should invest and have proposed investments in education and self-development. Others supported or criticized government bonds. For example, in the article from [aktualne.cz](#) [24], they wrote that Greece is proof that government bonds are not a risk-free investment, as textbooks say. It was also stated that if one longs for a bond fund, it should be the one with a short duration, preferably around three years. That is why government bonds analysis is not included in this thesis. With bonds, there is minimum, if any, investment for longer than 10 years and therefore does not meet the conditions for inclusion in this bachelor's thesis. Similarly, the commodities of rice and possibly other agricultural crops, which were again mentioned in the articles, could not be included. In this case, the most common way of trading is through futures, which would probably not be suitable for a model investor from this paper as well as it is, likewise to bonds, not ideal for long-term investment [37].

2.3 Data summary

Next, data for the five selected assets will be presented. For Invesco Global Water ETF (PIO), Gold, České energetické závody (CEZ) and Vanguard S&P 500 ETF (VOO), details are available in daily records. Therefore around 2800 records are publicly accessible for these instruments for the past ten years. Specifics of these assets are from Yahoo finance, where the day's opening, highest, lowest, and closing prices are available. For further analysis, this thesis works with closing prices [32].

Since this paper is based on a model investor who draws information from articles on the Internet that are easily accessible and understandable, the author decided to determine the method of investing in the above-selected investment instruments again according to the recommendations on the Internet. For investment in water, Invesco Global Water ETF (PIO) was selected. Recommendations on where exactly to invest in drinking water were not the easiest to find. Though PIO was, among some mentioned alternatives, pointed out the most. It was, for example, recommended on a page called investopedia.com [34], where they specifically suggested Invesco Global Water Portfolio ETF (PIO). An ETF is an exchange-traded fund and in this case, it is one that focuses on companies in the water industry. This specific ETF tracks Nasdaq OMX Global Water Index. This fund was also mentioned on some Czech websites, for example, on colosseum.cz [26]. Here it was mentioned that PIO is one of the best-known water ETFs. Others mentioned were IH20, PHO, and CGW. The issuer of the PIO fund is Invesco Ltd., which is a company that offers a range of mutual funds and exchange-traded funds [40]. The data for further analysis is obtained from finance.yahoo.com, which provides historical prices per one share of Invesco Global Water ETF in US dollars. Data for tracking these prices are available from 2007 to the present [23].

Purchase of investment bricks was chosen for investment in gold. According to an article on aktualne.cz [4], it is most advantageous to buy investment metal in the form of ingots, i.e. gold bars or in the form of investment coins. Since one gram of gold bar cost around a thousand Czech crowns (approximately 44 US dollars), it was suggested more advantageous to shop regularly and monitor market developments. It was not recommended to invest in gold-oriented securities because they can be practically worthless in a crisis. At the same time, they do not propose to invest in gold jewelry, as jewelry does not work with pure gold. The model optimizes how many troy ounces of gold, while one troy ounce is 31.10 g [20], will be included in the resulting portfolio. Historical gold prices are obtained from finance.yahoo.com [19] and these prices indicate how many US dollars one troy ounce of gold costs. Data is available for the years 2000–2022, i.e. to the present.

Next, for the investment in the CEZ, a.s. company, it was chosen to buy shares of the firm directly. They are found under the name CEZ.PR. As it is a Czech company, historical prices per share are given in Czech crowns. Data is again obtained from finance.yahoo.com [10] and, as with gold, it is available from 2000 to 2022.

After was the determination of the investment method in the S&P 500 index. As mentioned, investing will take the form of buying shares in a fund that follows this index. As with the water investment, the decision to select a particular fund was made based on internet articles and recommendations. On the website patria.cz [39], it is stated that direct purchases into ETFs, in this case, are most often made through Vanguard S&P 500 ETF (VOO) or SPDR S&P 500 ETF TRUST (SPY) funds. Then, the Vanguard S&P 500 ETF was mentioned again on the seznamzpravy.cz website [54], where it is directly stated in which Czech banks or portals it is possible to invest in it. Moreover, this particular fund was often mentioned on websites in English as well. For example, they describe two Vanguard funds, namely The Vanguard S&P 500 ETF, which is traded as a share, and then the Vanguard 500 Index Fund Admiral Shares, which is a mutual fund. The advantage of both of these variants is that they have meager fees and accurately copy the S&P 500 index results. However, since the Vanguard S&P 500 ETF was much more to be found in these articles as well as investing in ETF has been promoted, as it is to be more accessible as a stock exchange fund, an investment in the VOO fund has been chosen [39]. Historical price data was once more found on finance.yahoo.com [52], and it is available from 2010 to 2022.

The last resource to be analyzed in the portfolio optimization is agriculture land, also referred to as farmland. There have not been many recommendations on how to invest with this asset. Thus, the author considered either investing in farmland through funds or investing directly in agricultural land by purchasing a certain number of hectares. However, funds of this kind tradable by Czech citizens are relatively rare and a minimum deposit of CZK 1 000 000 was required in the few found. Such a fund is, for example, the Český fond půdy or the Agrární půdní fond [25]. For these reasons, it was decided to invest directly in agricultural land and to optimize the number of hectares purchased in the portfolio. However, complications subsequently arose with this investment. It was difficult to find source data for the calculations. This paper focuses on an investor from the Czech republic. Therefore the data needed to be for Czech land. Unfortunately, only yearly data were available. Their source is an agriculture land market report from June 2022 from a portal farmy.cz [59]. The historical prices are in Czech crowns and are available for the years 2005 to 2021.

Since the data must be comparable, all of the assets to be included in the portfolio had to be in yearly records. As well as they all have to be for the same years. Therefore for each instrument, there are 12 historical prices available for further analysis, specifically for the years 2010-2022. This means that the price of the last day of each year was selected for the four instruments, apart from farmland. Unfortunately, this will significantly reduce the amount of data, so it will be necessary to use special tests for small samples in the analysis. Additionally, a large amount of information is likely to be lost.

Data of prices for each asset are from the years 2010 to 2021 and, therefore, the returns are available from the year 2011. As an example, prices and returns in percentages for PIO are shown in tab. 2.1. Tables for other assets are shown in appendix A – Historical data. Annual returns are calculated as one year's price ratio to the previous year.

Table 2.1: Prices and returns of PIO [23]

Year	Price in USD	Return%
2010	20.01	
2011	15.59	-22.09
2012	18.01	15.52
2013	23.10	28.26
2014	22.96	-0.61
2015	20.93	-8.84
2016	20.80	-0.62
2017	26.04	25.19
2018	23.08	-11.37
2019	30.89	33.84
2020	34.93	13.08
2021	43.45	24.39

3. Methodology

Based on the literature research and information about data, this chapter will present the methodological procedures used in the empirical analysis. It is divided into four subsections, one with fitting distribution to data and second with simulations. In third, the risk assessment methods used in this thesis are discussed, and lastly, the model for portfolio optimization will be presented.

3.1 Probability distribution fitting

The identification of a suitable probability distribution can be verified mainly by means of a test on the distribution and a by graph. It is best to use more than one graph, as the division can fit the data only in some respects. As stated in Chapter 2, while analyzing the data for this thesis, it will be necessary to deal with an issue of a small data set. Therefore, testing normality or other probability distributions may produce biased results. The fewer data available to perform the test, the more likely it is that a false assumption about the goodness of fit of the distribution will be made.

Firstly, numeric tests will be discussed. The data are assumed to be continuous, and therefore the probability distribution is determined from the continuous ones. As discussed more fully in Chapter 4, the problem with the available data is not only its quantity but also the presence of negative values, which significantly narrows the options for choosing a distribution. In addition to the commonly known normal distribution, the data set with negative values can take on more complicated distributions, which this paper cannot cover. Hence, such cases will be tested for normal distributions, and if normality is not disproved, it will be assumed. Well known distributions that do not assume negative data are, for example, exponential, Weibull, lognormal, or gamma [56].

Normality will be tested first, as the normal distribution is considered the most basic one. As asymmetric distribution is suspected, which is confirmed in Chapter 4, as the data for every investment instrument are skewed, it is most recommended to use Anderson Darling or Shapiro-Wilk normality test. At the same time, the Shapiro-Wilk test is suitable for symmetric distributions as well and, therefore, can respond even to poorly analyzable data. Anderson Darling is in the category of empirical distribution tests on normality, whereas the more known Shapiro-Wilk is in the type of tests based on regression and correlation. The null hypothesis is set for both tests as the data is normally distributed, and the alternative hypothesis states non-normal data. The significance level is set to $\alpha = 0.05$ for further analysis. The Shapiro-Wilk test was initially limited to a sample size of at least 50, but it has been refined over time. The algorithm used in this paper is AS R94 11, for which a sample size of non-missing values is acceptable within $3 \leq n \leq 5000$. This algorithm is defined in the R language in the function specified directly for the Shapiro-Wilk test. Similarly based tests

but not used in this thesis due to assumptions, are the Shapiro-Francia and Ryan-Joiner tests. Anderson-Darling test compares hypothesized distribution with empirical distribution function. It is more computationally intensive than, for example, the similar Cramer-von Mises test but can handle smaller samples and places more emphasis on the tails of the distribution [38, 44, 57]. For testing in R studio, the function is under a package "nortest" and the sample size required is $n > 7$, while missing values are allowed [2]. Next, testing for other continuous distributions mentioned will be conducted in R for data of positive observations.

Secondly, the suitability of the probability distribution will be examined using graphs. The density plot and cumulative distribution function are most commonly used for the graphical fitting of a distribution. The density plot, referred to as a histogram, shows data on the x-axis as the probability of the data taking on a specific value. In this histogram, the y-axis represents the theoretical densities of the individual probability distributions. Thus, it is possible to compare whether the curves fit the shape of the histogram. A cumulative distribution function plot (CDF) compares the space under the probability density function of data against probability distribution. Again, the more the curves match for this graph, the more the theoretical distribution fits the data. The Q-Q chart, where Q-Q represents quantile-quantile, is created using quantiles (ordering the data values in order), which is produced by forming a graph from the quantiles of one distribution against the quantiles of another distribution. For the purposes of this thesis, the x-axis represents the theoretical quantiles of the distribution, and the y-axis represents the empirical quantiles. With the help of this graph, the similarity of scale and skewness between the data and a given distribution can be determined. Thus this graph can be considered a graphical way of performing goodness of fit findings. P-P plot, which stands for percentile-percentile, is a special type of Q-Q plot. Similar to the Q-Q plot, the x-axis stands for the theoretical data and theoretical probabilities, and the y-axis represents the empirical probabilities. The P-P graph plots the cumulative distribution functions that are compared with each other. The reference line on the chart is at 45° . If the curve is plotted on this line, it indicates equality between the tested values [11, 13].

3.2 Simulation

Using the Monte Carlo simulation, it is necessary to randomly generate needed values from the determined distributions. Specifically, this paper will involve the generation of return measures, for which the Inverse transform sampling is widely used, if a condition of continuous distributions is met. This method uses the relation $r = F(x)$ to determine a value from a given distribution that is equal to the value of the inverse distribution function of the distribution. The transformed formula is as follows: $x = F(r)^{-1}$. Where r is a randomly generated number standardly from uniform distribution $R(0,1)$, this can be applied to all mentioned distributions [49]. In this work, a simulation takes place in generating returns, from which the average return and risk measures are calculated. The second option would be to simulate

an optimization model and then average its result. However, the first mentioned approach was chosen due to the regular investment.

The simulation runs for p years, with a regular investment of q amount of money each year. For the first year of investment, the return is calculated according to the equation 3.1:

$$R_{ilj} = (1 + F(r)^{-1})q \quad i = 1, 2, \dots, n; \quad l = 1; \quad j = 1, 2, \dots, m. \quad (3.1)$$

where $F(r)^{-1}$ represents the random simulation from a distribution, R_{ilj} stands for expected value of investments of year l , where l for this formula only equals to one, while j represents an iteration from the simulation.

The simulated values of investments of the following years are shown in the formula 3.2, which additionally takes into account the return or loss of previous years while continuing to add the regular yearly investment and considering the random simulation from a distribution:

$$R_{ilj} = (1 + F(r)^{-1})(q + R_{i,l-1,j}) \quad i = 1, 2, \dots, n; \quad l = 2, 3, \dots, p; \quad j = 1, 2, \dots, m. \quad (3.2)$$

From these computations, one iteration of the Monte Carlo simulation is calculated as follows in 3.3:

$$R_{ij} = \frac{(\frac{R_{ipj}}{pq} - 1)}{q} \quad i = 1, 2, \dots, n; \quad j = 1, 2, \dots, m, \quad (3.3)$$

where R_{ij} denotes an average yearly return after 10 years of regular investment of i -th asset, while n stands for the total number of assets and j -th iteration, for which m shows the total number of iterations in a simulation. R_{ipj} represents the value of investments of i -th asset and the last, i.e., p -th, year considered and the j -th iteration. This return includes the regular investments of all p previous years.

m of such return simulations will subsequently be performed, resulting in m measures of return for a single investment instrument, from which an average simulated return of the asset can be calculated 3.4:

$$R_i = \frac{\sum_{j=1}^m R_{ij}}{m} \quad i = 1, 2, \dots, n, \quad (3.4)$$

where $R_{ij}(i = 1, 2, \dots, n; j = 1, 2, \dots, m)$ represents j -th expected return, where m is the total number of simulated observations, of the i -th asset, while n is the complete number of assets included [6, 36].

The total return of the portfolio, marked as R_p , is calculated as a weighted sum 3.5:

$$R_p = \sum_{i=1}^n R_i x_i, \quad (3.5)$$

where x_i represents the percentage of the portfolio invested in the i -th instrument and R_i is defined according to the calculation above.

3.3 Risk identification

As discussed in Chapter 1, there are many different approaches to risk. This thesis will focus on the risk identified by standard deviation, average absolute negative deviation, and probability of loss. The first two risk calculations focus on return or rather express concern about not reaching a high enough yield. This attitude is most common in portfolio analysis. Standard deviation is the most basic and the oldest method for calculating risk introduced, whereas the average absolute negative deviation is newer and more computationally and interpretatively demanding. However, it does not penalize positive deviations, and at the same time, it applies to the data used in this work. For these reasons, both approaches of this type are included. The third characteristic reflects the fear of financial loss. For some investors it may be essential that their investments do not fall in value, as this can lead to premature withdrawals and panic.

For all the following formulas, $r_i (i = 1, 2, \dots, n)$ represents risk of each financial instrument. Standard deviation is defined as shown in 3.6 [56]:

$$\sigma = \sqrt{D(X)} = \sqrt{E(X^2) - E(X)^2}, \quad (3.6)$$

for which $E(X)$ is an expected value of a random variable. In this case, the returns and $D(X)$ stand for such variable's variance. Therefore for purposes of this thesis, the formula would be as in 3.7, where vector R represents a vector of returns of assets:

$$r_i = \sqrt{D(R)} = \sqrt{E(R^2) - E(R)^2} \quad i = 1, 2, \dots, n. \quad (3.7)$$

Next, the concept of average absolute negative deviation is introduced in the equation 3.8:

$$r_i = \frac{\sum_{R_{ij} < \bar{R}_i} (\bar{R}_i - R_{ij})}{m^-} \quad i = 1, 2, \dots, n, \quad (3.8)$$

where $R_{ij} (i = 1, 2, \dots, n; j = 1, 2, \dots, m^-)$ considers simulated data as j -th simulated return, where m^- is a number of simulated observations below the average return, of the i -th asset,

were n is a total number of assets considered for a portfolio. Average return across all simulations of i -th asset is represented as $\bar{R}_i$.

Lastly, the probability of loss is defined in formula 3.9:

$$r_i = \frac{\#(R_{ij} < 0)}{m} \quad i = 1, 2, \dots, n, \quad (3.9)$$

for which the numerator $\#(R_{ij} < 0)$ represents the number of negative simulated returns, and m means the total number of simulated returns.

The total risk of the portfolio, marked as r_p , is calculated as a weighted sum 3.10:

$$r_p = \sum_{i=1}^n r_i x_i, \quad (3.10)$$

where the risk is calculated by one of the variants above. There is no quadratic form of variance in this work, meaning that covariance is not used, which could be expected for the measure of risk calculated through the standard deviation. This is because, firstly, linearity is preferred and, secondly, due to the lack of data and the portfolio's diversity, the dependencies between the variables are not considered necessary.

3.4 Portfolio optimization

The portfolio optimization model used in this thesis combines two models. One that maximizes returns 3.11 and second that minimizes risk 3.12 [8]:

$$\begin{aligned} MAX \quad z &= \sum_{i=1}^n x_i R_i, \\ \sum_{i=1}^n x_i &= 1, \\ x_i &\in X \quad i = 1, 2, \dots, n, \end{aligned} \quad (3.11)$$

$$\begin{aligned} MIN \quad z &= \sum_{i=1}^n x_i r_i, \\ \sum_{i=1}^n x_i &= 1, \\ x_i &\in X \quad i = 1, 2, \dots, n, \end{aligned} \quad (3.12)$$

where $x_i(i = 1, 2, \dots, n)$ indicates what percentage of the total portfolio was invested in each investment instrument. $R_i(i = 1, 2, \dots, n)$ represents average returns of each instrument and $r_i(i = 1, 2, \dots, n)$, as in risk identification, contains risks of the assets respectively. $\sum_{i=1}^n x_i = 1$ expresses that the individual percentage components of the portfolio must equal 100%, which means that all financial resources will be invested. Number of assets is defined by the variable n , which must be the same for all mentioned vectors. $x_i \in X$ for both models expresses any additional conditions on the portfolio elements allocation, most commonly the minimum investment and portfolio diversification [12, 32].

These two models are combined and used as a multi-criteria optimization model that simultaneously maximizes return and minimizes risk, as visible in equation 3.13:

$$\begin{aligned} \text{MAX} \quad z &= \sum_{i=1}^n x_i R_i - \sum_{i=1}^n x_i r_i, \\ \sum_{i=1}^n x_i &= 1, \\ x_i &\in X \quad i = 1, 2, \dots, n. \end{aligned} \tag{3.13}$$

The comparability of returns and risk can be a problem, since even if they have the same units (in this case percentages), they may differ in orders of magnitude. That is why it is necessary to standardize the variables. The approach to standardization shown bellow was chosen because thanks to it both variables, i.e., the rates of return and risk, take values from 0 to 1. Therefore it is possible to interpret where on this scale they are after the optimization process. 0 is taken as the worst possible situation for both variables and 1 as the best. The rate of return and risk in the optimization is standardized by their minimum and maximum, so using both extremes. This approach is also chosen because it is necessary to avoid the problem of negative values of criterion functions. If only one extreme value was used for standardization, which would be negative, then the expression of relative relations would be very complicated. This concept is inspired by discrete decision problems, where for example, the weighted sum approach (WSA) method is used. This method is based on constructing a linear utility function on a scale from 0 to 1 [28]. The equation for standardization of returns s_R is as in 3.14:

$$s_R = \frac{\sum_{i=1}^n x_i R_i - e^{\min}}{e^{\max} - e^{\min}}, \tag{3.14}$$

where $e^{\max}$ stands for the maximum possible portfolio return and $e^{\min}$ denotes the minimum portfolio return. The standardized risk measure s_r is calculated as in 3.15:

$$s_r = \frac{f^{\max} - \sum_{i=1}^n x_i r_i}{f^{\max} - f^{\min}}, \tag{3.15}$$

in which f^{max} denotes the maximum risk of the portfolio and f^{min} indicates the minimum risk. The highest achievable return e^{max} is obtained by maximizing returns as in equation 3.16 that also defines the highest possible risk f^{max} .

$$\begin{aligned} MAX \quad z &= \sum_{i=1}^n x_i R_i, \\ \sum_{i=1}^n x_i &= 1, \\ x_i &\in X \quad i = 1, 2, \dots, n. \end{aligned} \tag{3.16}$$

The resulting shares of investment instruments in the portfolio $\mathbf{x}^* = (x_1^*, x_2^*, \dots, x_n^*)^T$ obtained by the optimization function from 3.16 are then substituted into the equations for calculating e^{max} 3.17 and f^{max} 3.18:

$$\sum_{i=1}^n x_i^* R_i = e^{max}, \tag{3.17}$$

$$\sum_{i=1}^n x_i^* r_i = f^{max}. \tag{3.18}$$

On the other hand, minimization of risk shown in formula 3.19 will lead to the lowest achievable risk f^{min} and the lowest possible return e^{min} will also be determined.

$$\begin{aligned} MIN \quad z &= \sum_{i=1}^n x_i r_i, \\ \sum_{i=1}^n x_i &= 1, \\ x_i &\in X \quad i = 1, 2, \dots, n. \end{aligned} \tag{3.19}$$

This time the retrieved vector $\mathbf{x}^{**} = (x_1^{**}, x_2^{**}, \dots, x_n^{**})^T$ representing the shares of investment instruments in the portfolio from the optimization 3.19 is used to calculate e^{min} and f^{min} using the equations 3.20, 3.21:

$$\sum_{i=1}^n x_i^{**} R_i = e^{min}, \tag{3.20}$$

$$\sum_{i=1}^n x_i^{**} r_i = f^{min}. \quad (3.21)$$

Regarding equations 3.18 and 3.21, the extremes f^{max} and e^{min} are determined in the context of the possible values of the 3.16 and 3.19 criterion functions, respectively. This ensures that these extremes will not be set at unnecessarily pessimistic levels, i.e., levels that they cannot achieve with simultaneous multi-criteria optimization.

Finally, the resulting portfolio optimization model, derived from previous formulas, is defined in the equation 3.22:

$$\begin{aligned} MAX \quad & z = s_R + s_r, \\ & \sum_{i=1}^n x_i = 1, \\ & x_i \leq uy_i \quad i = 1, 2, \dots, n, \\ & x_i \geq vy_i \quad i = 1, 2, \dots, n, \\ & y_i \in \{0, 1\} \quad i = 1, 2, \dots, n, \end{aligned} \quad (3.22)$$

where the return R_i and risk r_i of each asset is the average of the simulated values, according to the previous subsection. Concerning the conditions for portfolio diversification and minimal investment, the minimum v and maximum u limit shares of one investment in the portfolio were determined. In both cases, the values of the variables are in the range 0 to 1. Binomial variables y_i indicate whether the i -th instrument was included in the portfolio or not. If it takes the value 1, it indicates the presence of the given instrument in the portfolio and the value 0 means the asset is not included.

4. Empirical research

The last chapter of this bachelor thesis will describe the portfolio optimization itself and everything that precedes it. The calculations are performed in three programs: R studio, Excel, and Lingo. R studio is used for graphs and normality verification, Excel for simulation, and Lingo, linked to data from Excel, for portfolio optimization.

4.1 Data preparation

First, an analysis of historical data was performed. For better comparability, charts have been created for all financial instruments based on annual data. At the same time, the maximum available amount of data was also used for an overview of price developments, i.e., for Invesco Global Water ETF (PIO), Gold, České energetické závody (CEZ) and Vanguard SP 500 ETF (VOO), with daily data and annual data for agricultural land. These additional graphs can be found in Annex 2–Distribution fitting and optimization models. The charts showing the development of price in the years 2010-2021 are plotted for each instrument separately, as CEZ shares and farmland hectares differ in units from the others. On the other hand, revenues are shown together in a chart since it is a dimensionless unit and is thus comparable between all financial instruments.

Figure 4.1 shows the development of the price of gold. It displays how much USD one troy ounce of gold costs. On this chart, we see a significant drop in price in 2013–2018. Conversely, high prices in other years, especially in 2012 and 2019. This corresponds to the statement in Chapter 2, where it was mentioned in the articles that the price of gold fluctuates according to world crises. When there is a crisis, whether a financial, pandemic or other one, investors buy gold and vice versa. In calm times, gold is undesirable. The price of gold responds to these fluctuations [1].

Next, Figure 4.2 shows the evolution of historical prices in USD per share of VOO, which follows the S&P 500 index. It is evident that this instrument has grown chiefly in the last 12 years. It peaked in the last year under review and was lowest in the first two years, with the lowest point in 2012.

After, Figure 4.3 shows the development of the share price of the Czech company CEZ. The price is quoted in Czech crowns per share. Interestingly, the curve reached its peak in 2021, but in 2012 the value of CEZ shares started to plummet and has not recovered for eight years. Suddenly in 2021, the price skyrocketed. The decline in 2012 is consistent with why recommendations to buy this stock were only to be found until this year and then disappeared completely, as mentioned in Chapter 2.

Lastly, Figure 4.4 depicts the historical prices of agricultural land. In this case, it is the price per hectare in Czech crowns. This investment instrument is interesting because it does not

fall once, which means that its value has increased at least a little every year. Although the growth of its prices is slower than the growth of, for example, VOO prices, it is also more stable. The highest point on the graph is again in 2021 and the lowest in 2010, so the first and last observed year respectively.

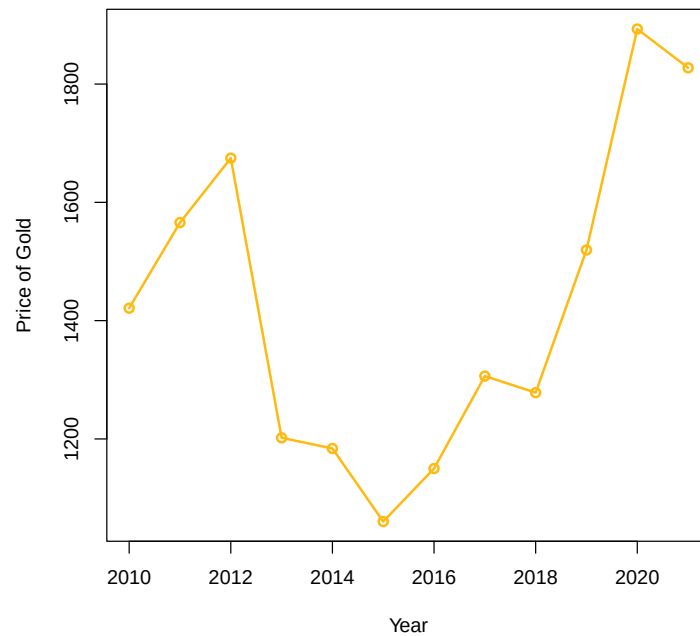


Figure 4.1: Gold price development in USD [19]

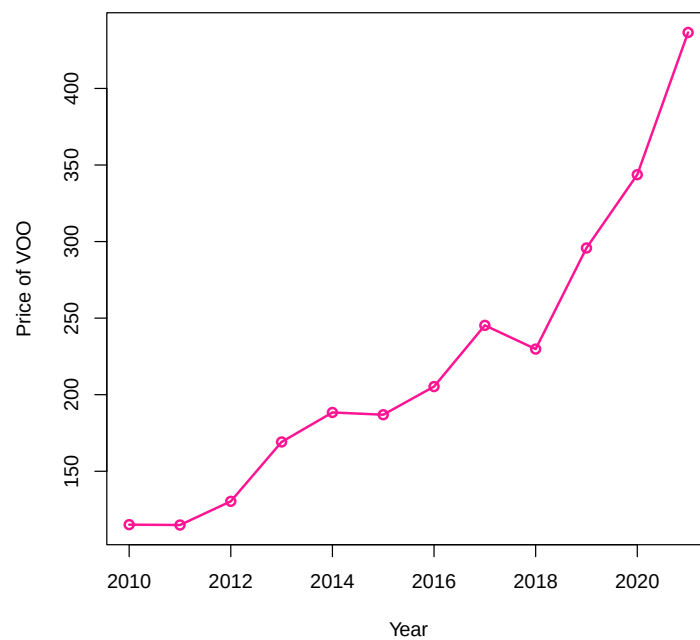


Figure 4.2: VOO price development in USD [52]

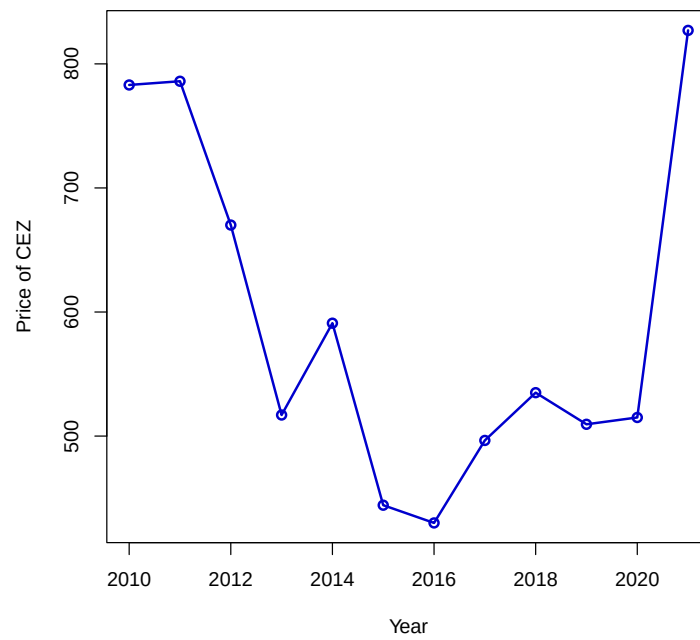


Figure 4.3: CEZ price development in CZK [10]

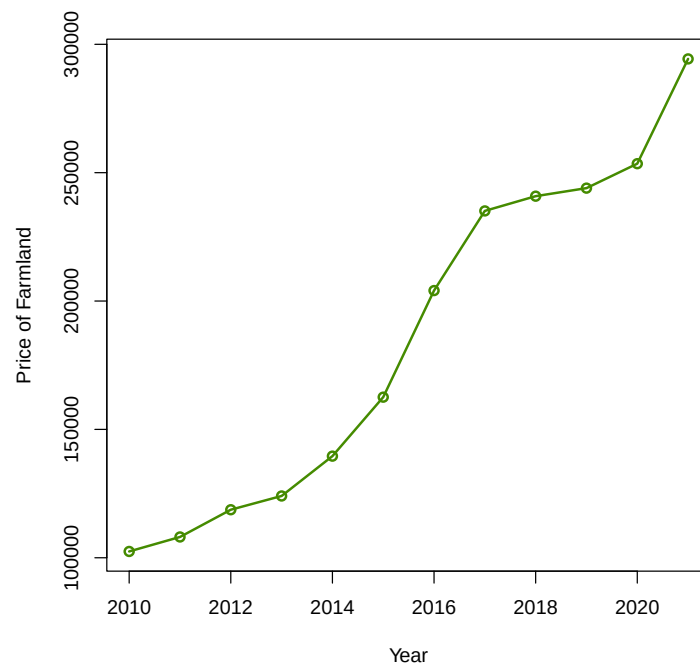


Figure 4.4: Farmland price development in CZK [59]

A percentage return for each year was calculated from historical prices, as mentioned in Chapter 2. The results were then averaged, and the average annual return was computed. At the same time, from the percentage returns of each year, the standard deviation was also calculated since it will be needed for the simulation assessment. Return as well as the standard deviation for each investment is depicted in Table 4.1.

Table 4.1: Yearly average return and standard deviation of each asset [10, 19, 23, 52, 59]

Name of asset	Return %	Standard deviation %
PIO	8.80	18.52
Gold	3.35	14.73
VOO	13.50	12.34
CEZ	2.65	23.42
Farmland	10.30	7.57



Figure 4.5: Returns in % [10, 19, 23, 52, 59]

Figure 4.5 shows the percentage returns of all investment instruments to be included in the optimization model. This chart is essential for comparing the fluctuation in returns between investments and also for tracking how much these returns are above zero. This threshold indicates how much an investment instrument appreciated or depreciated in value within the years 2011–2021. On the chart, it is possible to see that the yields of PIO and VOO have a similar curve, but PIO mainly reaches greater extremes. As noted in the analysis of historical prices, farmland yields have not fallen under zero even once, with a noticeable fastest price growth in 2016. Gold and CEZ shares are the most frequently below zero of all included instruments, which means they have lost value most often in the last ten years among the mentioned assets. At the same time, it is interesting to note that despite this,

CEZ has the highest recorded return among all in 2021. Gold generated the lowest measured return, or rather the highest price loss, in 2013.

4.2 Simulation

For Monte Carlo simulation, it is necessary to find the probability distribution of the historical data. Concretely, the distribution of revenues needs to be determined, as from these, the simulation will be carried out. As mentioned in Chapter 3, two tests will be used to verify normality. Specifically, the Anderson-Darling normality test and the Shapiro-Wilk normality test. These tests were conducted in R using functions:

```
ad.test()
shapiro.test()
```

For all investment components of the portfolio, the result was the impossibility to reject the null hypothesis, as can be seen in Table 4.2, and therefore normality can be assumed. This applies at a test significance level of 0.05, as determined in Chapter 3.

Table 4.2: Normality test

Name of asset	Anderson-Darling p-value	Shapiro-Wilk p-value
PIO	0.519	0.549
Gold	0.718	0.754
VOO	0.635	0.518
CEZ	0.143	0.084
Farmland	0.371	0.347

However, for reasons already mentioned in Chapter 3, it is also necessary to verify this claim using graphs and consider other distributions. The familiar lognormal, gamma, exponential, or Weibull distribution cannot be assumed for all investment instruments except agricultural land, as they take values below zero. Thus, tests for these distributions were only performed for the farmland investment. As can be seen in Table 4.3. In all these tests, the null hypothesis was not rejected even once. The highest p-value, i.e., the furthest from rejecting the null hypothesis, was in the test for the Weibull distribution. This statement holds true even when compared with the tests for normality. Thus, although the null hypothesis for normality tests was not rejected, the author considers it more appropriate to use the Weibull distribution for further analysis. Functions used for the analysis of the fit of these distributions:

```
lnorm_test()
gamma_test()
exp_test()
weibull_test()
```

Table 4.3: Distribution fitting to Farmland

Distribution	p-value
Lognormal	0.52
Gamma	0.65
Exponential	0.41
Weibull	0.98

This statement is also demonstrated in Figure 4.6, where we can see that the Weibull distribution fits the data better than the other distributions. Of course, it should be mentioned that even though it seems most suitable, the distribution does not copy the data exactly. This is again probably due to the small amount of data. The density and CDF plots are used because they can be considered essential goodness-of-fit plots. The other two graphs show the mismatch, namely, the Q-Q graph emphasizes the lack of fit distribution tails the P-P graph emphasizes the lack of fit in the middle of the distribution. Since the Q-Q plot is mainly "S" shaped, heavier tails of the data are assumed against most of the distributions. This statement is visible in the histogram as well [13]. Distribution fitting of the normal distribution to historical data of other instruments can be found in Appendix B – Distribution fitting and optimization models.

Once the probability distributions were determined, the actual simulations were performed. As recommended in Chapter 2, the model investor plans to invest regularly and over a longer time horizon. It was chosen to invest over a 10-year period as this roughly corresponds to the historical observation period. Fifty thousand Czech crowns per year are invested, which means that in 10 years, half a million Czech crowns will be invested in this portfolio.

It is known from Chapter 1 that this thesis treats returns as a probabilistic variable since it is never known what the future will look like. For this reason, the returns for the next ten years are random variables from a given distribution. For the normal distribution used for PIO, Gold, VOO, and CEZ, the parameters μ and σ^2 need to be known. These are calculated from historical data and are shown in Table 4.1, where μ stands for average return, and σ^2 is represented by variance, i.e., standard deviation. The probability is randomly generated. Therefore, to calculate probabilistic estimates of returns for example for the PIO instrument, this is the Excel command used:

```
=NORM.INV(RAND(),11.89,16.27),
```

For each year of investment, the return is calculated accordingly, resulting in 10 generated returns. The percentage return is then multiplied by the regular investment amount in the first year and in all the following years by the value of the previous year's investment plus the value of the regular investment, i.e., 50 000. This is shown in Chapter 3 in formulas 3.1 and 3.2. The final amount that the investor should achieve over the investment period is obtained. In this case, it is the five hundred thousand Czech crowns invested plus

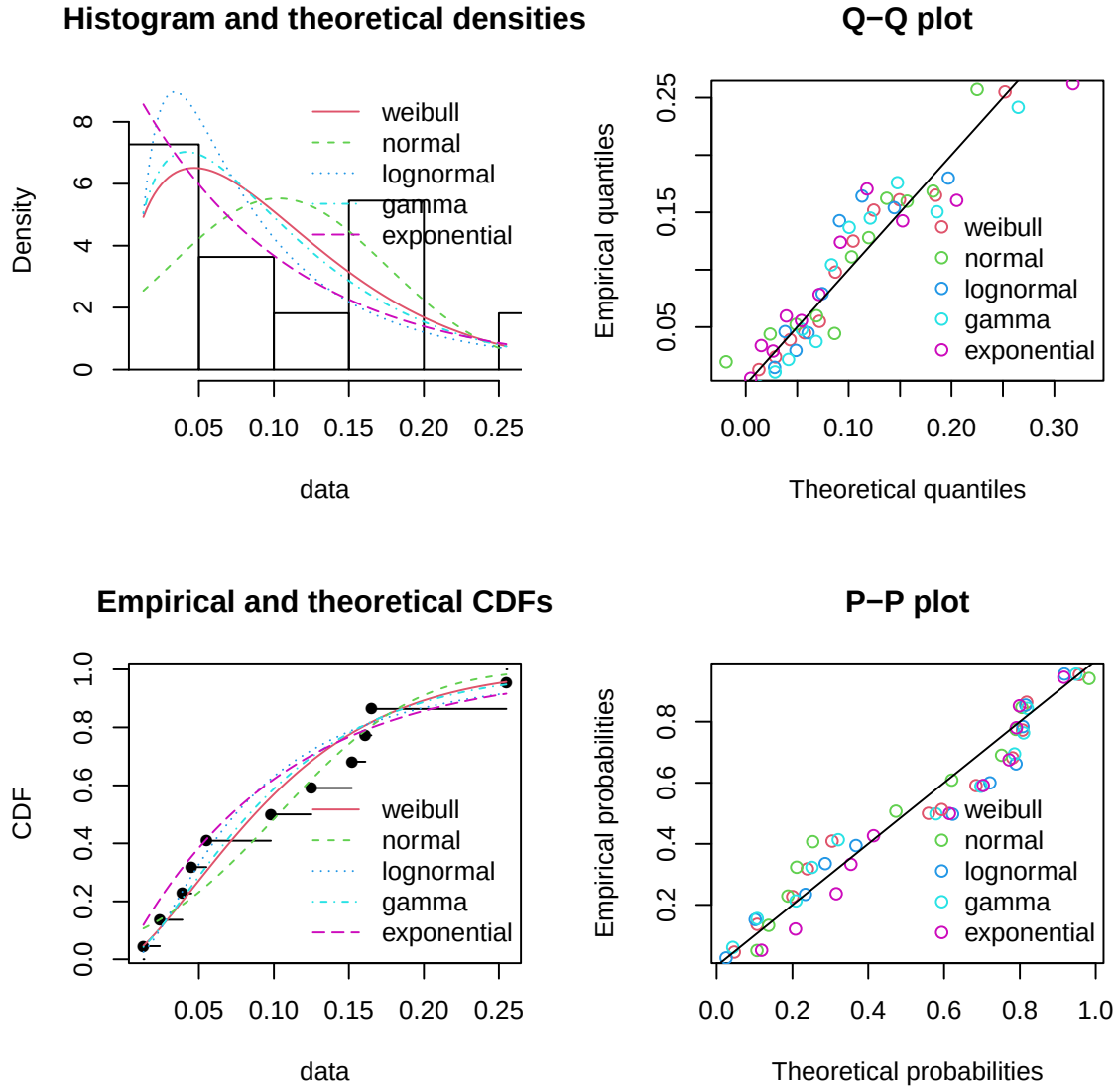


Figure 4.6: Farmland distribution fitting

the profit or loss over the ten-year period. From this value, the annual percentage return over these ten years is calculated according to formula 3.3. Afterwards this value is generated m times, where m is set as thousand, thus creating the simulation discussed in the previous chapters. Specifically, ten return measures are generated thousands of times through a probability function that affects the evolution of investments over a ten-year period. From these the final return after the investment period and average yearly return is calculated each time. The simulation is performed using a data table in Excel, which falls under the what-if analysis [6]. This produces a thousand average annual returns, which are then averaged to give the average annual return after the simulation, the procedure is shown in equation 3.4. The simulated values are also used to calculate risk. There are several approaches to risk to consider. As determined in Chapter 3, three ways of calculating risk are included in this thesis. Namely, the risk calculated as standard deviation (3.7), average absolute negative

deviation (3.8), and probability of loss (3.9). Results of return and risk computations after the simulation are shown in Table 4.4.

According to the previous analysis, as far as agricultural land investment is concerned, it was decided to use the Weibull distribution. Therefore, the procedure for generating returns is slightly different. The first step was to specify the distribution parameters, which were obtained in R-studio. The parameters k , which reflects the shape, and λ , which indicates the scale, were found using the function [13]:

`fitdist()`

Since it is not possible to use the inverse Weibull distribution function in Excel, it was necessary to follow the formula from Chapter 3. Precisely for this distribution, the equation is specified by the formula 4.1:

$$x = \lambda \sqrt[k]{-\ln(1 - r)}, \quad (4.1)$$

which was derived according to the inverse transformation method discussed in Chapter 3. from the formula 4.2 [43]:

$$F(x, k, \lambda) = 1 - e^{-(x/\lambda)^k}. \quad (4.2)$$

The parameter k equals 1.41 and λ equals 0.11, with r being randomly generated. The next computation procedures concerning the regular investments and simulations are the same as for the normal distribution.

Table 4.4: Return and risk of each asset after Monte Carlo simulation [59]

Asset name	Return %	Standard deviation %	Average absolute negative deviation %	Probability of loss %
PIO	6.46	6.26	4.22	12.40
Gold	2.21	3.51	2.49	29.20
VOO	11.35	5.04	3.66	0.10
CEZ	1.48	5.71	3.68	45.50
Farmland	7.67	2.64	1.79	0.00

The results in Table 4.4 show that the highest return is expected from the investment instrument following the S&P 500 index. On the contrary, the lowest returns are expected from the investment in CEZ. Interestingly, the risk calculated for gold using the standard deviation and the average absolute negative deviation is lower than for most other investments. On the other hand, the risk is much higher when examining the probability of loss. For purposes of this thesis, for the calculations above $n = 5$, as five investment instruments are discussed.

4.3 Portfolio optimization

By simply looking at the resulting values of returns and risks, it is possible to assume a more significant amount of invested funds going to VOO and less to CEZ shares, but the ideal way of distribution of funds is not apparent. For this reason, portfolio optimization was conducted. Script is written for the Lingo program, which is connected to Excel with the results from previous calculations. In accordance with Chapter 3, maximum u and minimum v investments have been determined. The author decided to allow a maximum of $u = 0.4$ of the portfolio to be invested into one asset. This way, the portfolio will have to include at least three instruments, and therefore, since the selection is made out of five possibilities, it could be considered a diversified portfolio. Additionally, when using 40%, it is assumed that differences between probabilities will more likely correspond to the preferences of assets. If, for example, the maximum investment of $1/3$ of the portfolio to one asset were used, then the optimization model would give each included instrument the same weight in the portfolio. Next, the minimum investment to include the asset in the portfolio is decided to be $v = 0.15$. The minimum investment is determined due to the fact that, although it has often been written in the articles analyzed in Chapter 2 that even an investment of 1000 Czech crowns counts, it is not possible in all cases to invest only such a small amount of money. For example, when buying agricultural land, such a small amount would probably not be sufficient. Therefore, if the investment is not advantageous by at least 15%, it will not be included in the optimized portfolio.

Three models were created to optimize the portfolio for all three types of risk computation, discussed in Chapter 3, from the simulated values. The model compiled for risk calculation as standard deviation is shown below, the other two models can be found in Appendix B—Distribution fitting and optimization models. Variables e and f stand for the sum of return or risk of each asset multiplied by the allocated percentage, respectively. Subsequently, the variables g and h representing their standardized values are defined in Chapter 3 as s_R and s_r . The following optimization model corresponds to the model 3.22.

```

MODEL:
SETS:
instrument/1..5/: risk, return, percentage, y;
ENDSETS

max = g + h;
e=@sum(instrument(i):return(i)*percentage(i));
f= @sum(instrument(i):risk(i)*percentage(i));
g = (e-6.253)/(8.957-6.253);
h = (4.362-f)/(4.362-3.45);
@sum(instrument(i): percentage(i)) = 1;
@FOR(instrument(i):(percentage(i)) >= 0.15*y(i));
@FOR(instrument(i):(percentage(i)) <= 0.4*y(i));
@FOR(instrument(i):@BIN(y(i)));

DATA:
risk, return = @ole('model.xlsx');
@ole('model.xlsx')=percentage,a,e,f,g,h,y;

ENDDATA
END

```

For risk calculated as the probability of loss, the optimal return-maximizing option has the same resulting portfolio composition as the risk-minimizing option. Therefore, the model for optimizing this portfolio is not standardized and it follows the 3.13 theoretical model.

The results of all three optimizations are shown in Table 4.5. As expected, investment in VOO shares, which follows the S&P 500 index, is always included in the optimized portfolio. On the contrary, investing in CEZ shares is not even once recommended, due to the lowest expected return between possible investments and the highest or second-highest risk value, according to risk identification. However, it is interesting that the agricultural land is included in the portfolio for each of the three approaches. The purchase of gold or POO shares is chosen as the third investment instrument. The distribution of funds into individual assets is also different in each case. Only the investment in farmland is always evaluated as advantageous up to the maximum allowed.

The approach to risk as a standard deviation from returns recommends that 40% of the funds in the optimized portfolio should be invested in agricultural land, 40% allocated to VOO shares, and the remaining 20% is reserved for investments in gold. The standardized approach to return and risk, specifically the values of the parameters g and h in the model above, determines where the given variable is in the interval from 0 to 1. In both cases, 0 means the worst possible result and 1 the best possible one. For the first measure of risk, both variables have similar results: standardized return is $g = 0.690$, and standardized risk is $h = 0.612$. The benefit of the two values is higher than 50%, thus, it could be argued that these are favorable values. This means that both values are closer to the best possible value.

Table 4.5: Optimized solution regarding risk selection

Name of asset	Portfolio share according to risk selection		
	Standard deviation	Average absolute negative deviation	Risk of loss
PIO	0	0	0.2
Gold	0.2	0.4	0
VOO	0.4	0.2	0.4
CEZ	0	0	0
Farmland	0.4	0.4	0.4

Secondly, the risk was analyzed in the form of average absolute negative deviation, which focuses only on negative deviations in the simulated data. In this case, it was recommended to invest 40% in farmland, 40% in gold, and 20% in VOO as the optimal portfolio. The ratio of invested funds in gold and VOO thus switched, compared to the previous approach. The standardized level of return for this case is $g = 0.958$, and the standardized risk level is $h = 0.427$. It is evident that this time there are significant differences within these variables. While the return is almost at its best value, the risk is below 50%, which means it is closer to the worst possible value. This is probably due to the fact that by taking into account only the negative deviation, the risk level for all instruments decreased. For this reason, there is a smaller difference in the calculated risks than in the previous approach. Therefore, as the differences between yields remain unchanged, the standardization generates a more considerable difference between the worst and best value. That is why the best profit value is more prioritized, as the negative impact of the less profitable composition of the portfolio would be more pronounced than a risk-prone portfolio.

Finally, the portfolio taking into account the risk of loss, was analyzed. In this case, it may be misleading that this risk came out of the simulations as zero for two variables. The investment in VOO and agricultural land is thus very advantageous in the optimization model compared to other assets. This approach recommended investing 40% of the funds in farmland, 40% in VOO and 20% in POO shares. Interestingly, it is impossible to evaluate the variables g and h , as the best possible return was achieved with the same portfolio composition as the best-acquired risk. This means that if the investor fears losses, by such a distribution of investments, one reaches the highest overall return among the selected instruments and, at the same time, the overall lowest risk.

The total return and risk percentages for each risk approach are included in the table 4.6. The highest expected return was calculated for the risk taken as probability of loss, as well as it produced the lowest expected risk. On the contrary, the lowest return is generated while taking into account only negative deviations. The highest risk is found for the risk taken as a standard deviation, which could have been predicted in advance, as it takes into account the most fluctuations of all three approaches.

Table 4.6: Total return and risk of the portfolio regarding risk selection

Total measures	Risk selection		
	Standard deviation	Average absolute negative deviation	Risk of loss
Return %	8.12	6.25	8.96
Risk %	3.80	2.49	2.44

To summarize, if an investor opts for the most stable investment, in which one does not want too much fluctuation in returns, one should follow a portfolio where risk is taken as a standard deviation. Thus, one would invest equally 40% of the allocated money in farmland and VOO shares and then 20% in gold. If the investor requires the lowest possible fluctuations only in the negative direction, one should focus on a portfolio where risk is identified as average absolute negative deviation. Again, one would invest 40% in farmland, but unlike the previous approach, one would only put 20% in VOO shares and the remaining 40% in gold. The third option, a portfolio based on risk as a probability of loss, should be chosen if the investor fears a loss, i.e., that the rate of return will turn negative. For this case, one should invest 40% of the funds in agricultural land, 40% in VOO shares, and 20% in PIO shares. To identify their priorities, investors should go through an investment questionnaire to reveal their approach to risk [55].

Last but not least, this chapter includes a reflection on Chapter 2, where the recommendations from Internet articles were discussed. Land, real estate, or farmland was indeed highlighted very frequently, so in this case, it can be determined that the recommendations were correct. Another big part of all the portfolios mentioned above is the investment in the S&P 500 index, or rather in a fund that this index follows. However, this way of investing in internet portals was rather criticized, so it can be said that authors and experts were wrong in this case. Recommendations for investing in water were less available than, for example, buying gold, which corresponds to the representation in the optimized portfolios. The fifth instrument selected was an investment in the shares of the Czech company CEZ. This opinion occurred very often until 2012, and in the following years, this investment was neither recommended nor criticized. Investment into CEZ shares was not included even once it optimized portfolios, so expert suggestions by the year 2012 genuinely differ from calculation-based recommendations.

Conclusion

To summarize, this thesis focused on investment decision-making based on suggestions from Internet news sources. The analysis of these recommendations itself was conducted as well as literary research on the investment topic and portfolio optimization. Based on the information collected, a portfolio has been compiled. Since each investor has different preferences, various approaches to risk were included in the thesis. In addition, the uncertainty aspect was also embedded since the future has to be considered unpredictable.

As the articles gave diverse advice, it was first necessary to decide on the overall investment approach. Therefore, a regular money deposit and a long-term investment horizon were chosen, concretely an investment planned for ten years with regular annual deposits. Subsequently, it was decided to invest in five instruments, i.e., in PIO shares, which focus on water, gold, VOO shares, which follow the S&P 500 index, shares of the Czech company ČEZ and agricultural land. As these are very diverse instruments that not only trade in different currencies and units but are either equity investments or outright commodity purchases, it was challenging to find relevant data and a suitable model. The portfolio optimization generated results for three types of approaches to risk. The optimal allocation of funds always included investment in farmland and VOO. The purchase of gold was selected as the third investment instrument for the two kinds of risk approaches, focusing on return fluctuations. For the risk approach in the form of fear of loss, investment in PIO shares was chosen instead of gold.

In addition to determining the ideal distribution of finances among the different assets, this work aimed to assess the accuracy of the claims made in the articles in the online portals. It could be argued that the recommendations were not explicitly wrong. Gold was the most frequently suggested asset in stable income-oriented portfolios. Land has the highest representation across the portfolios, which is consistent with the recommendations. Water investment, which was one of the surprising investment proposals for the author, despite the rise in PIO share prices over the years, was notched into the portfolio only when considering the probability of loss. The author expected the water investment to be more included based on comparing price trends. Investments in CEZ shares were much recommended before the severe fall in price. Logically, the instrument was not praised much after that, but an unpleasant discovery probably followed for those who, based on the suggestions from the articles, bought shares before 2012. Interestingly, investing in stocks of a fund that tracks the S&P 500 index was hardly recommended in the online newspapers. Actually, it was rather criticized. This asset was included in the analysis mostly for comparison purposes, as this is how the index is often used in practice. However, it has a place in all three optimized portfolios, and so it can be argued that the experts' recommendations did not pay off here. At the same time, it is worth noting that if one had invested in any of the instruments included in this paper in 2010, there would be an appreciation in savings for all five of them. However, in the case of an investment in CEZ, for example, the return would probably hardly cover inflation.

Based on the information above, the author believes that it may be beneficial to read the investment recommendations of internet portals, but it is not necessarily worth following them. They are undoubtedly helpful for a general overview and to get an idea of what it is possible to invest in. For example, an investment in water or mentioned investment in rice, which for the reasons listed in Chapter 2 could not be included in the analysis in this thesis, could be considered unconventional and might not have occurred to the investor without reading about it. At the same time, however, the articles often lack a more detailed view of investing in the recommended instruments and what to look out for. Thus, it could easily happen that an uninformed investor would choose a fund with unnecessarily high fees or buy gold without a certificate, and so on.

With all these conclusions, it must be considered that this thesis could not capture all essential information for a deeper analysis. Due to the fact that it is a heterogeneous portfolio, variability is achieved but with that, problems arise. Such as, the data amount from four instruments had to be significantly eliminated. This fact has led to the problem of determining the probability distribution, the accuracy of which would increase with more data. In order to make the instruments comparable, only long-term investments in which money can be invested regularly had to be taken into account. Some other recommendations, such as the rice mentioned above or government bonds, could not be included because of the investment horizon, while they could have been an interesting way of allocating money. Another major limitation of this work is the failure to take into account returns from holding the asset, such as rent in the case of land and dividends for stocks. These returns could, of course, disadvantage investment into gold, from which no such return is derived. On the other hand, costs have also not been included, neither one-off charges when buying or selling nor regular expenses. In the case of funds, these could be management fees or administration fees. The problems are listed here as pitfalls and limitations of this thesis, but at the same time, they can be taken as suggestions for further research.

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Appendices

A. Historical data

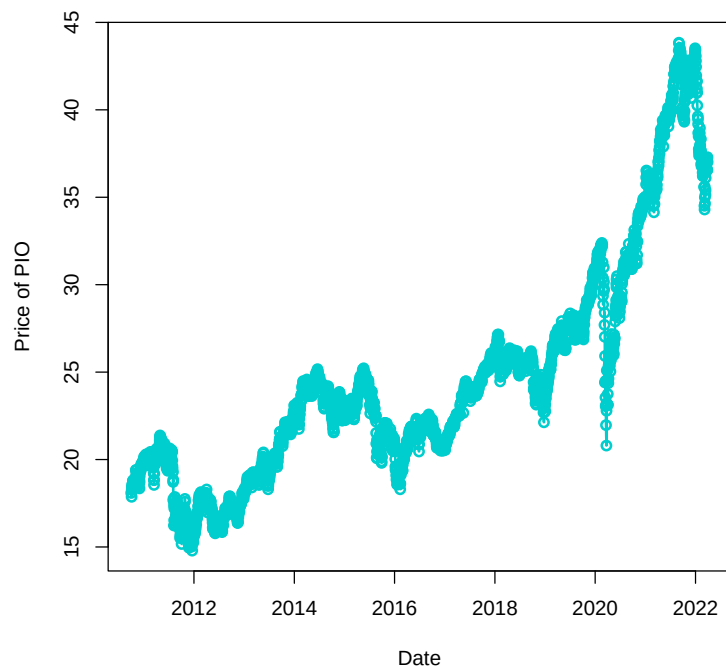


Figure A.1: PIO price development in days (USD)

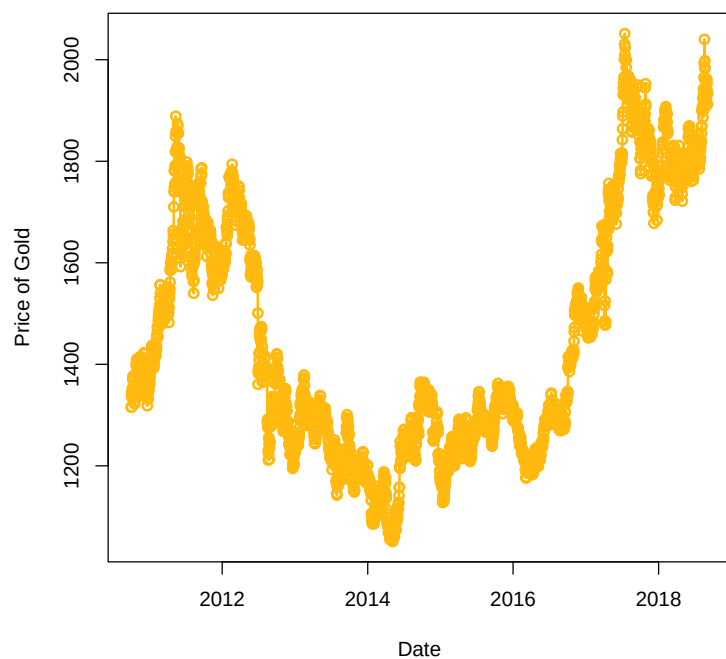


Figure A.2: Gold price development in days (USD)

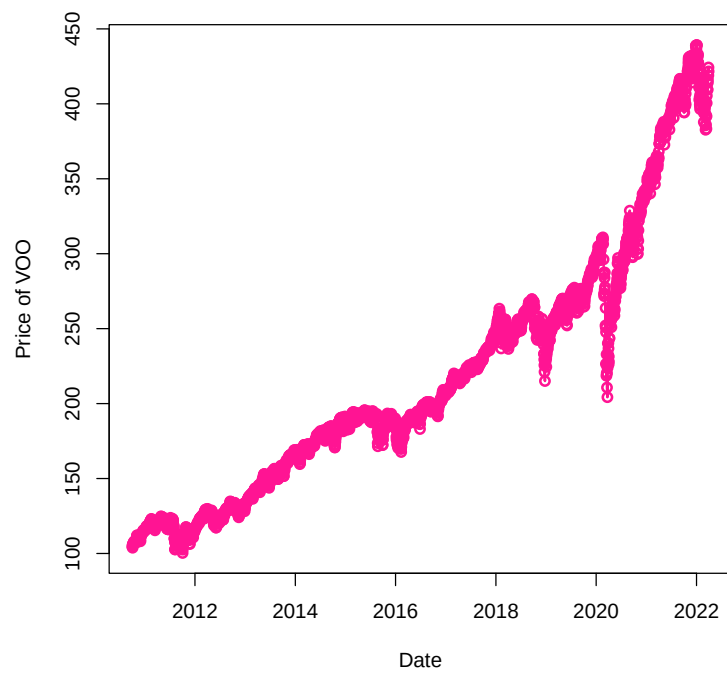


Figure A.3: VOO price development in days (USD)

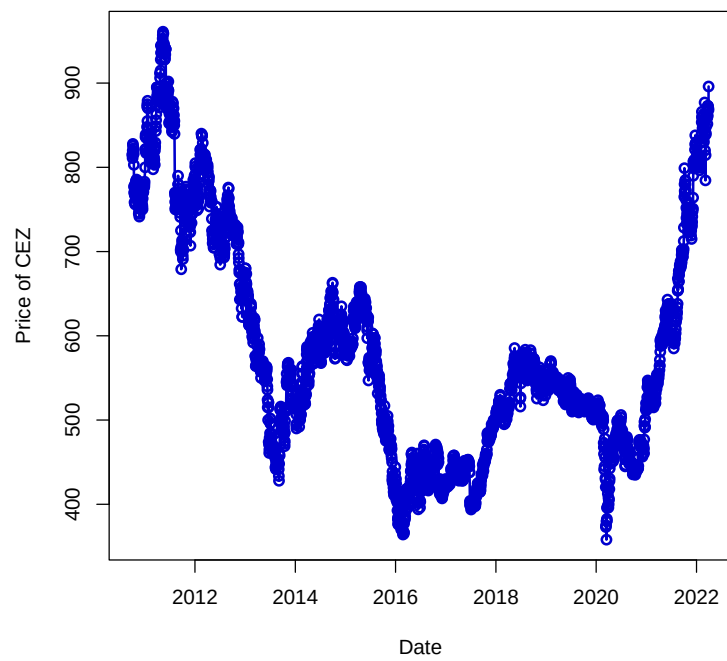


Figure A.4: CEZ price development in days (CZK)

Table A.1: Prices and returns of Gold [19]

Year	Price in USD	Return%
2010	1421,10	
2011	1565,80	10,18
2012	1674,80	6,96
2013	1201,90	−28,24
2014	1183,90	−1,50
2015	1060,30	−10,44
2016	1150,00	8,46
2017	1306,30	13,59
2018	1278,30	−2,14
2019	1519,50	18,87
2020	1893,10	24,59
2021	1827,50	−3,47

Table A.2: Prices and returns of VOO [52]

Year	Price in USD	Return%
2010	115,14	
2011	114,90	−0,21
2012	130,38	13,47
2013	169,15	29,74
2014	188,40	11,38
2015	186,93	−0,78
2016	205,31	9,83
2017	245,29	19,47
2018	229,81	−6,31
2019	295,80	28,72
2020	343,69	16,19
2021	436,57	27,02

Table A.3: Prices and returns of CEZ [10]

Year	Price in CZK	Return%
2010	783,00	
2011	786,00	0,38
2012	670,10	−14,75
2013	517,00	−22,85
2014	591,00	14,31
2015	444,30	−24,82
2016	430,00	−3,22
2017	496,50	15,47
2018	535,00	7,75
2019	509,50	−4,77
2020	515,00	1,08
2021	827,00	60,58

Table A.4: Prices and returns of Farmland [59]

Year	Price in CZK	Return%
2010	102456,00	
2011	108100,00	5,51
2012	118712,00	9,82
2013	124070,00	4,51
2014	139590,00	12,51
2015	162565,00	16,46
2016	204085,00	25,54
2017	235111,00	15,20
2018	240850,00	2,44
2019	243985,00	1,30
2020	253510,00	3,90
2021	294326,00	16,10

B. Distribution fitting and optimization models

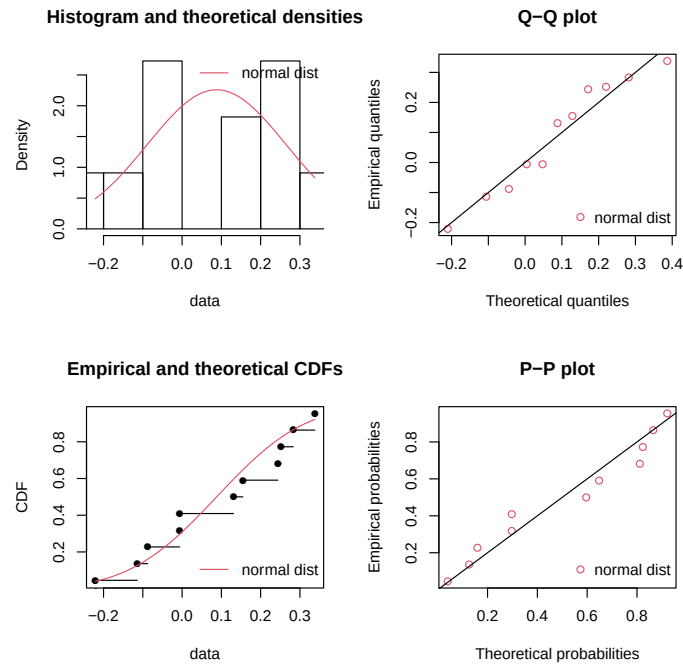


Figure B.1: PIO distribution fitting

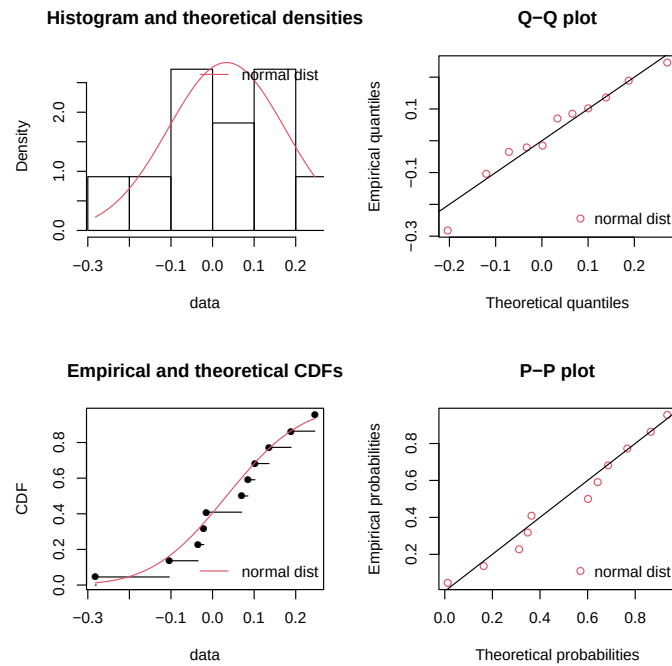


Figure B.2: Gold distribution fitting

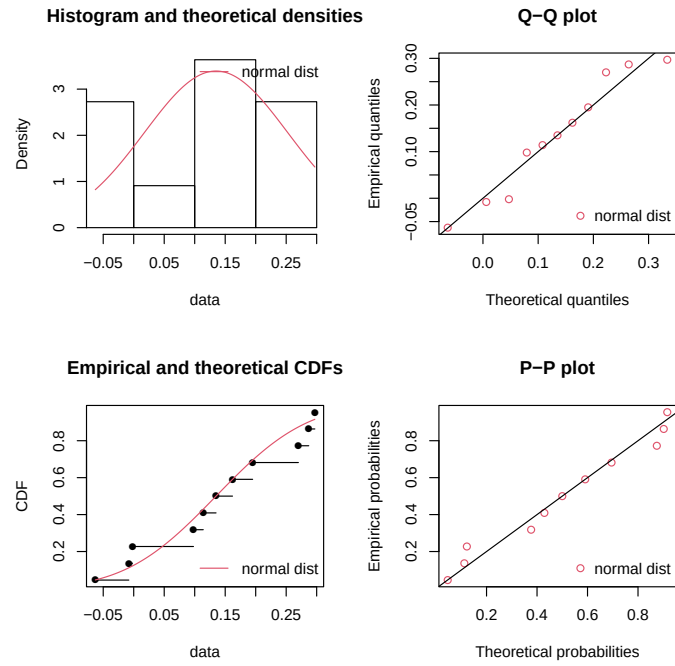


Figure B.3: VOO distribution fitting

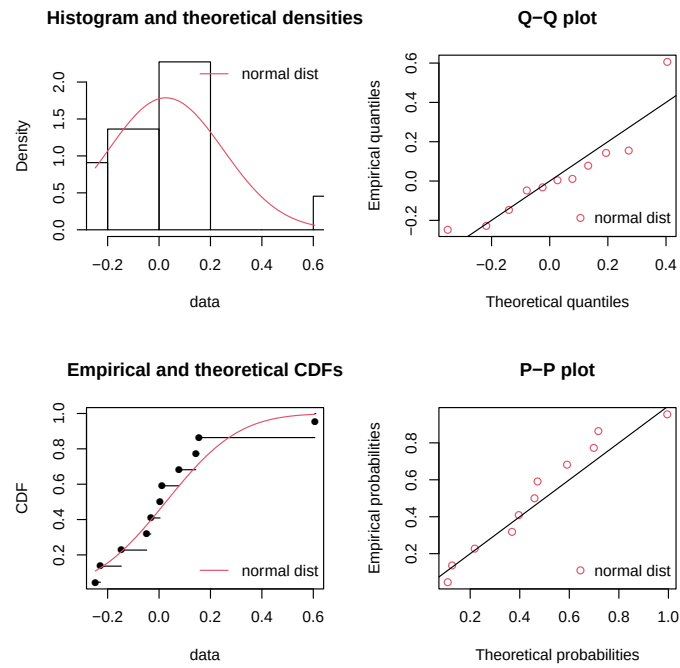


Figure B.4: CEZ distribution fitting

```

MODEL:

SETS:
instrument/1..5/: risk, return, percentage, y;
ENDSETS

max = g + h;
e=@sum(instrument(i):return(i)*percentage(i));
f= @sum(instrument(i):risk(i)*percentage(i));
g = (e-4.234)/(8.957-4.234);
h = (3.04-f)/(3.04-2.462);
@sum(instrument(i): percentage(i)) = 1;
@FOR(instrument(i):(percentage(i)) >= 0.15*y(i));
@FOR(instrument(i):(percentage(i)) <= 0.4*y(i));
@FOR(instrument(i):@BIN(y(i)));

DATA:
risk, return = @ole('model.xlsx');
@ole('model.xlsx')=percentage,e,f,y,g,h;

ENDDATA

END

```

```

MODEL:

SETS:
instrument/1..5/: risk, return, percentage, y;
ENDSETS

max = e-f;
e=@sum(instrument(i):return(i)*percentage(i));
f= @sum(instrument(i):risk(i)*percentage(i));
@sum(instrument(i): percentage(i)) = 1;
@FOR(instrument(i):(percentage(i)) >= 0.15*y(i));
@FOR(instrument(i):(percentage(i)) <= 0.4*y(i));
@FOR(instrument(i):@BIN(y(i)));

DATA:
risk, return = @ole('model_bak.xlsx');
@ole('model_bak.xlsx')=percentage,e,f,y;

ENDDATA

END

```